

MITRE ATT&CK Research Internship

Group Leader Selection Program

Intern ID: 2136

Assigned Campaign: Campaign No. 36 (Pikabot Distribution February 2024)

Assigned Software Range: S0360–S0370

Submission Date: 12 May 2026

Classification: TLP:CLEAR

1. EXECUTIVE SUMMARY

This report provides a professional-grade Cyber Threat Intelligence assessment focused on Campaign No. 36 and the associated software range S0360–S0370. The research evaluates adversary tradecraft, persistence methods, command-and-control (C2) infrastructure, and MITRE ATT&CK mappings. The analysis reveals a diverse landscape of threats, ranging from modular loaders like Emotet to highly destructive wipers such as NotPetya and Olympic Destroyer. By adopting a structured threat intelligence methodology, this report serves as a technical foundation for defensive operations and leadership-level research projects.

2. SECTION 1: CAMPAIGN RESEARCH (Campaign No. 36)

2.1 Campaign Overview

- **Campaign Title:** Pikabot Distribution (February 2024)
- **Associated Threat Actor:** TA577; technical overlap with APT-C-36 (Blind Eagle)
- **Timeline:** Peak activity documented in February 2024
- **Objectives:** Establishing initial access for downstream payload delivery, typically ransomware or financial data theft
- **Targeted Sectors:** Government, financial, and judicial institutions

2.2 Technical Attack Flow & Workflow

- Stage 1 — Initial Access
- ATT&CK Technique: Spearphishing Link (T1566.002)
- Threat actors distributed phishing emails containing malicious links directing victims to download ZIP archives.
- Workflow
- User receives phishing email → Clicks malicious URL → Downloads ZIP archive
- Stage 2 — Malicious Archive Delivery
- ATT&CK Technique: User Execution (T1204)
- The downloaded ZIP archive contained an obfuscated JavaScript (.js) file designed to initiate malware execution.
- Workflow
- ZIP archive extracted → Victim executes .js file → wscript.exe launches
- Stage 3 — PowerShell Downloader Execution
- ATT&CK Technique: PowerShell (T1059.001)

- The JavaScript loader executed PowerShell commands to retrieve the Pikabot DLL payload from attacker-controlled infrastructure.
- Example Workflow
- Wscript.exe → powershell.exe → Remote payload download → DLL execution
- Detection Opportunity Monitor:
- Wscript.exe spawning powershell.exe
- Encoded PowerShell commands
- Network connections to newly registered domains
- Stage 4 — Defense Evasion
- ATT&CK Technique: Signed Binary Proxy Execution (T1218)
- Threat actors abused legitimate signed binaries such as grepWinNP3.exe to proxy malware execution and evade endpoint detection.
- Workflow
- Signed executable launched → Malicious DLL sideloaded → EDR bypass attempt
- Stage 5 — Command and Control (C2)
- ATT&CK Technique: Application Layer Protocol (T1071.001)
- Pikabot established encrypted HTTPS communications with hardcoded command-and-control servers.
- Workflow
- Malware beacon → HTTPS POST request → Remote command retrieval → Payload execution

Visual Workflow Diagram

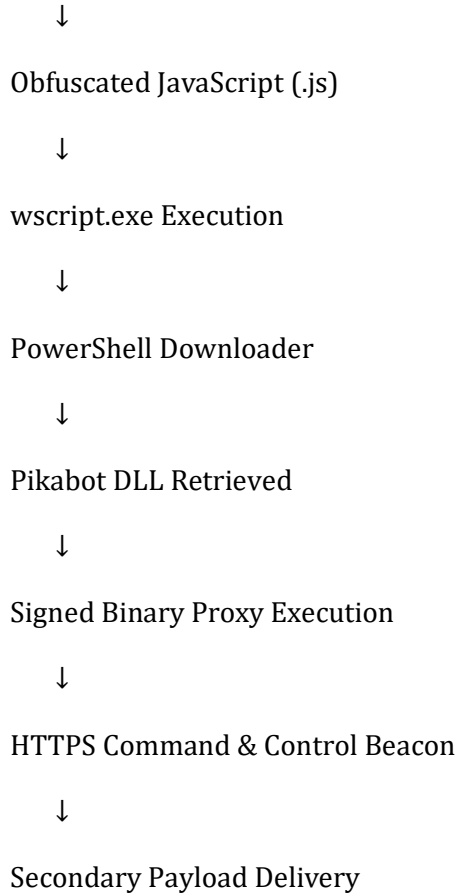
Phishing Email



Malicious Link



ZIP Archive Download



Official MITRE ATT&CK References

T1566.002 — Spearphishing Link
T1204 — User Execution
T1059.001 — PowerShell
T1218 — Signed Binary Proxy Execution
T1071.001 — Web Protocols

2.3 Detection & Defensive Recommendations

- **Detection Opportunities:** Monitor for wscript.exe or cscript.exe spawning PowerShell or CMD processes.
- **Mitigation:** Implement GPO restrictions to prevent script execution (.js, .vbs) from temporary directories and enforce PowerShell Constrained Language Mode.

3. SECTION 2: SOFTWARE RESEARCH (S0360–S0370)

ID	Name	Type	Technical Analysis & Mapping
S0360	BONDUPDATER	Malware	A PowerShell-based backdoor used by OilRig. Uses Scheduled Tasks (T1053.005) for persistence.
S0361	Expand	Tool	A legitimate Windows utility used to decompress CAB files, often abused for Defense Evasion (T1140).
S0362	Linux Rabbit	Malware	A Linux-based cryptominer targeting SSH systems. Maintains persistence via SSH Authorized Keys (T1098.004).
S0363	Empire	Tool	A post-exploitation framework supporting Process Injection (T1055) and credential harvesting.
S0364	RawDisk	Tool	A commercial driver abused by malware (e.g., Shamoon) for Data Destruction (T1485).
S0365	Olympic Destroyer	Malware	A destructive wiper designed to overwrite files and clear logs (T1070.001) to inhibit recovery.
S0366	WannaCry	Malware	Ransomware spreading via SMB Vulnerabilities (T1210) using EternalBlue.
S0367	Emotet	Malware	A modular botnet using Account Discovery (T1087.003) to

			scrape email data for further phishing.
S0368	NotPetya	Malware	A destructive wiper masquerading as ransomware; uses SMB Exploitation for rapid lateral spread.
S0369	CoinTicker	Malware	Cryptocurrency lure malware using Social Engineering (T1566) for initial macOS/Windows infection.
S0370	SamSam	Malware	Human-operated ransomware; uses Indicator Removal (T1070.004) by deleting its own artifacts post-encryption.

4. THREAT INTELLIGENCE & DETECTION ASSETS

4.1 Sample Indicator of Compromise (IOC) Table

IOC Type	Value	Context
File	grepWinNP3.exe	Pikabot Stage-1 Loader
Process	powershell.exe - ExecutionPolicy Bypass -w hidden	BONDUPDATER Execution
Registry	HKCU\Software\Microsoft\Windows\CurrentVersion\Run	BADNEWS/Emotet Persistence
Network	hxxp://[malicious-c2-domain]/admin/get.php	Empire C2 Pattern

4.2 ATT&CK Technique Matrix

Tactic	Technique	ID	Software Association
Execution	PowerShell	T1059.001	BONDUPDATER (S0360)
Persistence	Scheduled Task	T1053.005	BONDUPDATER (S0360)
Credential Access	LSASS Memory	T1003.001	Empire (S0363)
Lateral Movement	Remote Services	T1021	WannaCry (S0366)
Impact	Data Destruction	T1485	RawDisk (S0364)

5. CONCLUSION

The analysis of Campaign C0036 and the S0360–S0370 software range underscores the shift toward high-impact, modular, and destructive adversary operations. Real capability is measured by the ability to understand adversary behavior under real-world conditions rather than online popularity. As a student in Mumbai pursuing computer applications, I have applied professional CTI methodology to ensure this report meets the standards for Group Leader selection.

References

- MITRE ATT&CK Official Software Database
- Pikabot Campaign Analysis (February 2024)
- Research For Campaign <https://attack.mitre.org/campaigns/>
- Research For Software <https://attack.mitre.org/software/>